I will now finish this and next time we will present the family of spatial interaction models and finally show how we can couple these together to build more general models: LUTI models

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Monday, 19 January 2026

Urban Simulation 2b

Spatial Interaction: The Family of Spatiaal Interaction Models

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Monday, 3rd March 2026

Modifying the Generic Model: The Family of Spatial Interaction

At Elsa's instigation, In this talk, I have annnotated the mass variables in the gravity model – O_i^{α} and D_j^{γ} as coefficients $T_{ij} = KO_i^{\alpha} D_j^{\gamma} \exp(-\beta c_{ij})$. This is because the extended model is used in the practical but also because it is more general. But I will now we will use $T_{ij} = KO_i D_j \, exp\left(-\beta c_{ij}\right)$ when it comes to the family of gravity models. With different constraints, it is tricky and laborious to notate the origin and destination variables with parameters, so I have dropped these

The Family of Spatial Interaction Models

We already implicitly indicated how we can constrain the model to produce specific types that meet various constraints on the amount of trips attracted to different locations. Our first and most basic model is unconstrained in that we ensure that the only constraints the model meets are on the total trips which are predetermined, that is $T = \sum_{i=1}^{n} \sum_{j=1}^{m} T_{ij}$.

Now if we know the trips attracted to the origin, or to the destination, or to both, then these give rise to different models. Including the basic model there are four in all that meet the following. This is the family.

1. The Unconstrained Model

$T_{ij} = KO_iD_j \exp(-\beta c_{ij})$ subject to $\sum_{i=1}^{n}\sum_{j=1}^{m} T_{ij} = T$

2. The Singly-Constrained Models

2a: The Origin- Constrained Model

$T_{ij} = A_iO_iD_j \exp(-\beta c_{ij})$ subject to $\sum_{j=1}^{m} T_{ij} = O_i$

2b: The Destination-Constrained Model

$T_{ij} = O_iB_jD_j \exp(-\beta c_{ij})$ subject to $\sum_{i=1}^{n} T_{ij} = D_j$

3. The Doubly Constrained Model

$T_{ij} = A_iO_iB_jD_j \exp(-\beta c_{ij})$ st $\sum_{j=1}^{m} T_{ij} = O_i$ & $\sum_{i=1}^{n} T_{ij} = D_j$

We can work out the full models by substituting the model on the left into the constraint on the right

Now these models not only distribute trips but if we add their flows over origins or destinations or both, the different models give us predictions of what happens at origins and destinations. These are then **location models** and their predictions are in red below

The Models	**The Predictions**
$T'_{ij} = KO_iD_j \exp(-\beta c_{ij})$;	$\sum_{j=1}^{m} T'_{ij} = O'_i$; $\sum_{i=1}^{n} T'_{ij} = D'_j$
$T'_{ij} = A_iO_iD_j \exp(-\beta c_{ij})$;	$\sum_{j=1}^{m} T'_{ij} = O_i$; $\sum_{i=1}^{n} T'_{ij} = D'_j$
$T'_{ij} = O_iB_jD_j \exp(-\beta c_{ij})$;	$\sum_{j=1}^{m} T'_{ij} = O'_i$; $\sum_{i=1}^{n} T'_{ij} = D_j$
$T'_{ij} = A_iO_iB_jD_j \exp(-\beta c_{ij})$;	$\sum_{j=1}^{m} T'_{ij} = O_i$; $\sum_{i=1}^{n} T'_{ij} = D_j$

It is laborious to work through all the models but we can pick one of these models and show how we can unpack it and work out the scaling constants. We will take the ***origin-constrained model*** and adapt it to a retailing example. Here is the model

$$T'_{ij} = A_i O_i D_j \exp\left(-\beta c_{ij}\right) \; subject\ to \; \sum_{j=1}^{m} T'_{ij} = O_i$$

and this model predicts what we want to know about where people will locate at the destination

$$\sum_{i=1}^{n} T'_{ij} = D'_j$$

Now we must figure out the scaling constant A_i. We substitute the model into $\sum_{j=1}^{m} T'_{ij} = O_i$ and re-arrange as

$$\sum_{j=1}^{m} T'_{ij} = A_i O_i \sum_j D_j \exp\left(-\beta c_{ij}\right) = O_i$$

and then by cancelling O_i from both sides we get

$$A_i = 1/\sum_j D_j \exp\left(-\beta c_{ij}\right)$$

This then lets us predict the amount of activity attracted to destination j

$$D'_j = \sum_{i=1}^{n} T'_{ij}$$

If the origin activity is expenditure on retailing where the population lives and the destination is some measure of the size of activity such as retail floorspace. Then the predicted activity is sales at the destination shopping centres

Now the new computer program on the web site

Spatial-Interaction-Models.ipynb

Lets you choose one of these four models and using the same hypothetical data set shown earlier , it runs the model and presents the input and output data and the correlations of predicted v observed trips, origins and destinations.

```
Enter the Grid Size: The Number of Squares Along One Side of a Square Grid, 5 or 10, say ---  5
The Size of the Hypothetical Spatial System is 5 Zones by 5 Zones, Making 25 in All

Enter the Parameter Value on Distance - beta - that You Think Best Fits the Data: It should be greater than 0 and less than 1 ----  0.5

You now have to input one of the Four Model Variants that you want to run

We will number these as

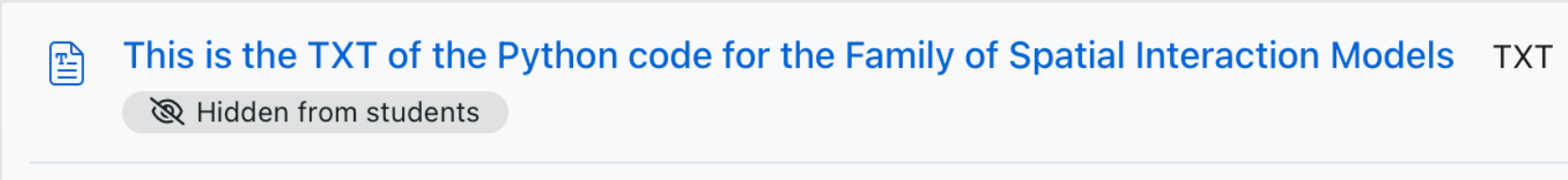
1 The Unconstrained Model as in the Basic Gravity Model Program
2 The Singly-Constrained (Origin-Constrained) Model
3 The Singly-Constrained (Destination-Constrained) Model
4 The Doubly-Constrained (Origin-and-Destination Constrained) Model

Type in the Relevant Number from 1 to 4 3
You have chosen the Destination-Constrained Model
Zone     ObsO    ObsD    PredO   PredD

1        28.93   34.57   22.18   34.57
2        45.28   32.50   41.04   32.50
3        39.63   39.23   37.97   39.23
```

I will now load Anaconda and then go to Moodle and get the text file and copy that to the clipboard and past it into a new file on the Jupyter Notebook. I will then run the model. Note that we can pick any of the four models from this program and run them and look at the predicted origins and destinations and trips as we did before for the unconstrained model.

You can download the ipynb file – which is a TXT file and paste it into the Jupyter Notebook and run it as I will show here



Urban Models: Coupled Spatial Interaction

Ok – we have the building blocks of more comprehensive urban models now. Imagine we want to build a model of where people work, where they live and where they shop.

We first define where people work as employment at origins E_i and where they live as population at destinations P_j. This can be modelled as singly constrained model where we predict P_j from E_i and then we predict where these people will shop which relates to the number of workers or size of the shopping centre S_i at the origins.

We thus start with journey to work model defined as

